

An Overview of Some Basic Robot Hardware

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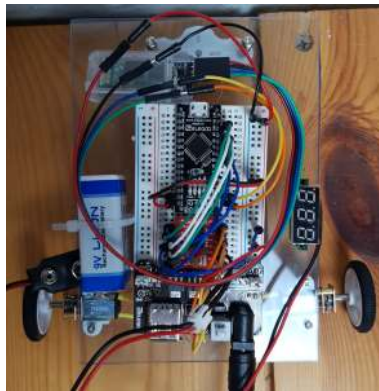
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Robótica y Percepción Computacional
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In this lecture I would like to talk about some basic kinds of robot hardware:

- Microcontrollers
- Actuators: DC Motors
- Effectors: Wheels
- Sensors: Absolute Encoder
- Sensors: Range
- Sensors: Video
- Communication: Bluetooth, WIFI
- Power: Batteries



Microcontrollers

Although there are many options, the Arduino family is very popular. (Strictly speaking an Arduino is not a microcontroller but rather contains a microcontroller.)

“Classic” Arduino Uno R3:

- Microcontroller (MCU): ATmega328P (8-bit AVR) running at 16MHz
- 32KB Flash Memory (program space, including .5K for the boot loader), 2KB SRAM (run-time data) 1KB EEPROM (long term storage)
- 14 digital I/O pins (6 of 14 with PWM) and 6 analog input pins
- USB-B and power jack (7-12 DC)
- 68.6mm x 53.4mm
- female headers



Image from https://commons.wikimedia.org/wiki/File:A000066_featured_4.jpg

Microcontrollers

Arduino Nano, almost the same as the Uno except:

- 22 digital I/O pins (6 of 22 with PWM) and 8 analog input pins
- USB mini-B (no power jack)
- 45mm x 18mm
- either male headers or without headers



Arduino Mega R3:

- ATmega2560 (8-bit AVR) running at 16MHz
- 256KB Flash Memory, 8KB SRAM, 4KB EEPROM
- 54 digital I/O pins (15 of 54 with PWM), 16 analog input pins, 4 UART
- USB-B and power jack (7-12 DC)
- 101.5mm x 53.3mm
- female headers



Images from https://commons.wikimedia.org/wiki/File:Arduino_nano.jpg and https://commons.wikimedia.org/wiki/File:Arduino_Mega.jpg

Microcontrollers

Arduino Uno R4 in both Minima and WIFI versions:

	R3	R4 Minima
Processor	ATmega328P	Arm Cortex M4
Speed	16 MHz	48 MHz
RAM	2K	32K
Flash	32K	256K
USB	USB-B	USB-C
Max V	12V	24V
Interfaces	I2C, SPI, UART	+ CAN
Other		analog output
Price (official)	€29.30	€22.00



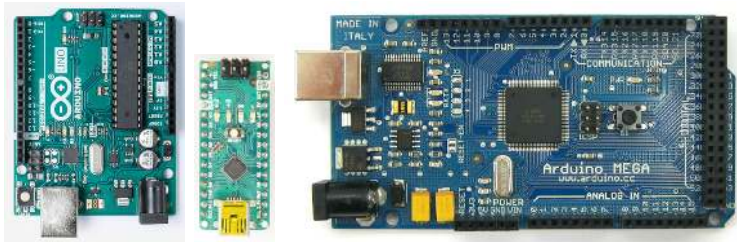
Arduino Uno R4 WIFI adds: ESP32-S3 (Wi-Fi and Bluetooth LE), 12x8 LED Matrix, Qwiic connector, ... €30.50

Images from https://commons.wikimedia.org/wiki/File:Arduino_Uno_R4_Minima.webp,
https://commons.wikimedia.org/wiki/File:Arduino_Uno_R4_WiFi.webp

Microcontrollers

For a simple robotics project any of those boards should work. (If you run out of pins you can always setup a bus.)

The Arduino is open hardware so compatible boards should work well and are generally cheaper.



Images from https://commons.wikimedia.org/wiki/File:A000066_featured_4.jpg,
https://commons.wikimedia.org/wiki/File:Arduino_nano.jpg, and
https://commons.wikimedia.org/wiki/File:Arduino_Mega.jpg

Some other options: ESP32 and Raspberry Pi (really a computer)



Images from

[https://commons.wikimedia.org/wiki/File:ESP32_Espressif_ESP-WR00M-32_Dev_Board_\(2\).jpg](https://commons.wikimedia.org/wiki/File:ESP32_Espressif_ESP-WR00M-32_Dev_Board_(2).jpg) and

https://commons.wikimedia.org/wiki/File:Raspberry_Pi_B%2B_top.jpg

Microcontrollers

Arduino UNO Q 2GB (€47.58) and 4GB (€64.66)

- Microprocessor (MPU): Qualcomm Dragonwing QRB2210, Linux capable
- Microcontroller (MCU): STM32U585
- RAM: 2/4 GB LLPDDR4
- Storage: 16/32 GB eMMC built-in
- Dual-band Wi-Fi 5 (2.4/5 GHz), Bluetooth 5.1
- High speed and classic UNO headers
- 8x13 LED matrix
- USB-C and Qwiic connectors
- Compatible with other UNO products



Images from https://upload.wikimedia.org/wikipedia/commons/5/5c/Arduino_UNO_Q.webp

Actuators: DC Motors

Many robots use DC motors as their actuators, this is because:

- they can be very simple and inexpensive to manufacture,
- they are available in many sizes with different properties,
- they can be very simple to control, and
- they work well with different kinds of batteries

We will talk about a number of basic kinds of DC motors:

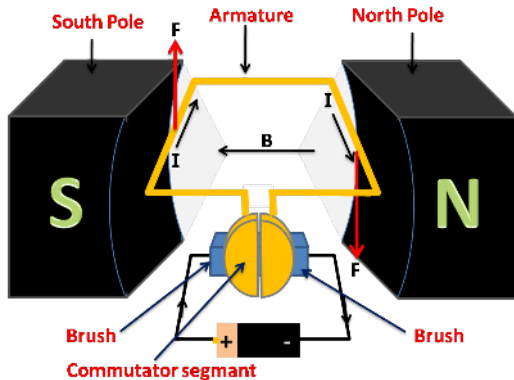
- Brushed
- Brushless
- Servo
- Stepper



Image from <https://create.arduino.cc/projecthub/muhammedazhar/make-your-first-arduino-robot-the-best-beginners-guide-d21a4f>

Actuators: Brushed DC Motor

Electric motors convert electrical energy into mechanical energy. One of the most commonly used electric motors is the brushed DC motor (simple two pole):

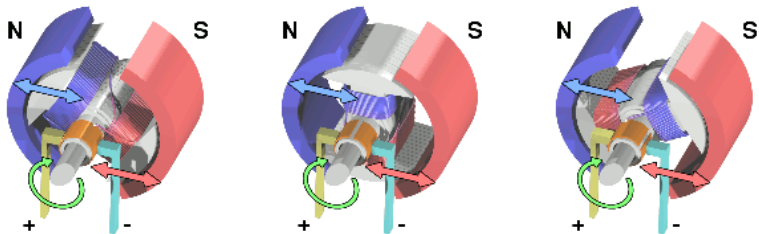


- brush - escobillas
- commutator - conmutador
- commutator plate - delgas
- armature/rotor - armadura/rotor
- coil - bobina
- stator - estator

Image from https://commons.wikimedia.org/wiki/File:Electric_motor_working_process.png

Actuators: Brushed DC Motor

Simple two pole brushed DC motor:



Images from http://en.wikipedia.org/wiki/Brushed_DC_electric_motor

Actuators: Brushed DC Motor

Three pole brushed DC motor

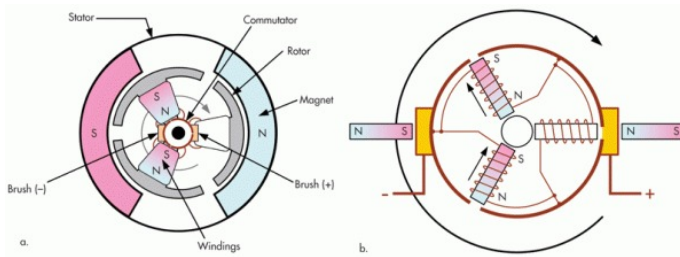


Image from <http://electronicdesign.com/analog/motor-drivers-advance-consumer-products>

Actuators: Brushed DC Motor

A brushed DC motor can be a very economical actuator for a robot (“hobby-type” motors are about €1).

Brushed DC motor control:

- The voltage sent to the motor controls the speed
- The polarity controls the direction

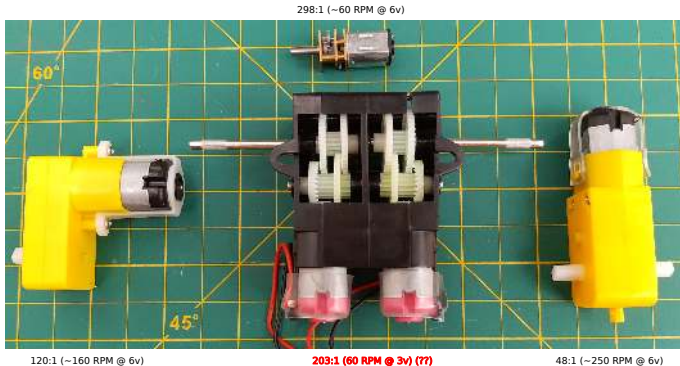


Images from https://commons.wikimedia.org/wiki/File:3V_DC_Motor.jpg and
https://commons.wikimedia.org/wiki/File:Small_DC_Motor_Rotor.JPG

Actuators: Brushed DC Motor

Brushed DC motors usually have high RPM and low torque. The “hobby motor” shown before produces about 4000 RPM at 6V but with very little torque.

In robots, gears are generally used to reduce the RPM and increase the torque of the motor.



Actuators: Brushed DC Motor

Pulse Width Modulation (PWM) is often used to control the motor's speed. It allows us to change the average voltage going to a motor:

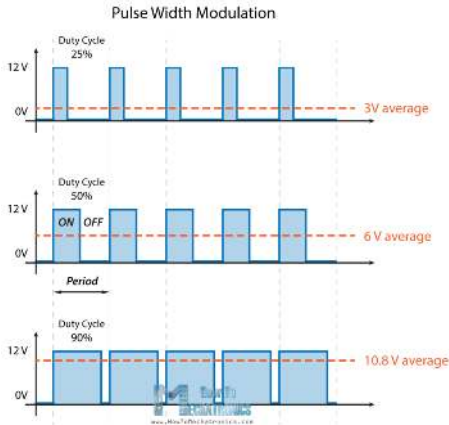


Image from <https://howtomechatronics.com/tutorials/arduino/arduino-dc-motor-control-tutorial-1298n-pwm-h-bridge/>

[//howtomechatronics.com/tutorials/arduino/arduino-dc-motor-control-tutorial-1298n-pwm-h-bridge/](https://howtomechatronics.com/tutorials/arduino/arduino-dc-motor-control-tutorial-1298n-pwm-h-bridge/)

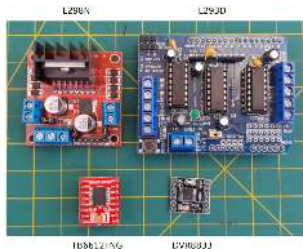
Actuators: Brushed DC Motor Driver

Often a motor driver is used to separate the power used by the motor (higher voltage, higher amperage) from the power used by the circuitry (often 3.3v or 5v and low current).

A typical TT motor works on 3-6v and can consume 150mA without a load and 1.5A when stalled. The efficiency of the chip is relevant when working with batteries. The below can vary, check your data-sheets!

Note, the L298N and the TB6612FNG are easily interchangeable.

	Peek A/Ch	Vout	Effic.(%)
L293D	1.2	4.5-12	34
L298N	2	5-35	40
TB6612FNG	3.2	2.2-13.5	81
DRV8833	2	2.7-10.8	~80
DRV8835	1.5	0-11	~80



Actuators: Brushed DC Motor Driver

Most drivers have h-bridges to control the direction of the motor(s). The L298N (dual h-bridge):

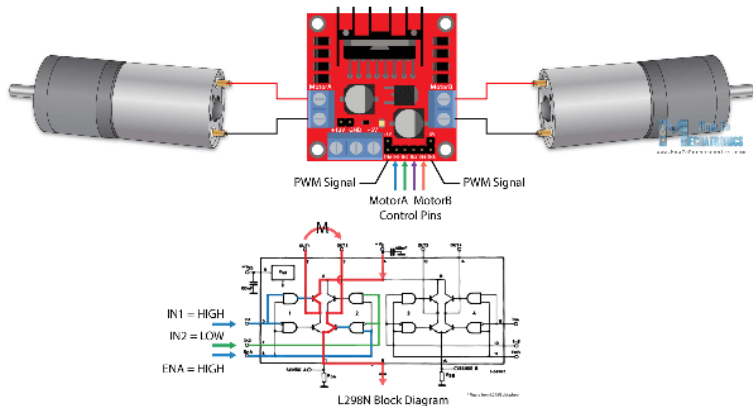


Image from <https://howtomechatronics.com/tutorials/arduino/arduino-dc-motor-control-tutorial-l298n-pwm-h-bridge/>

Actuators: Brushless DC motor

Brushless DC motor (turn the brushed motor inside out, substitute brushes with electronic control, the left image is of a computer cooling fan):

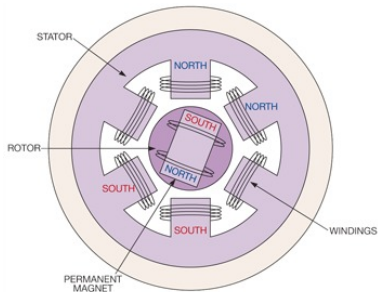
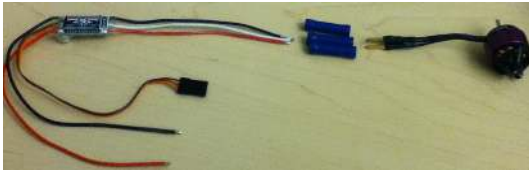


Figure 1 BLDC motors have a rotor with a permanent magnet containing north and south poles. The stator comprises multiple electromagnets.

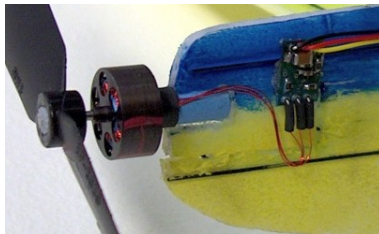
Images from <http://en.wikipedia.org/wiki/File:Poles.jpg> and <http://www.edn.com/design/systems-design/4363599/Hardware-controlled-brushless-dc-motors-ease-the-burden-on-CPUs>

Actuators: Brushless DC motor

A brushless motor with an electronic speed control module (ESC)



A M-60-5-3 DC brushless motor and a BLC-1 controller.



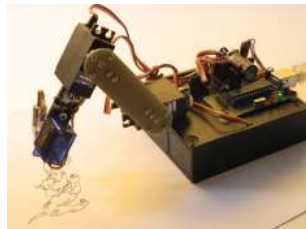
Images from https://commons.wikimedia.org/wiki/File:Horizontal_Exploded_View_of_Thunderbird_9_ESC_Butt_Connectors_and_Motor.JPG and <https://commons.wikimedia.org/wiki/File:Brushless-Motor-DUM60.jpg>

Actuators: Servomotor

Servomotors are not a 'class' of motor but rather a motor with feedback control.

Often, servomotors are used to position parts of a robot. For example to control the angles of a robotic arm.

They constantly compare the current shaft position/velocity to the required shaft position/velocity and use the error to correct (closed-loop control).



Images from https://commons.wikimedia.org/wiki/File:Theremino_Robot_Arm.jpg and <http://www.polytechnichub.com/difference-between-ac-servo-motor-and-dc-servo-motor/>

Actuators: Servomotor

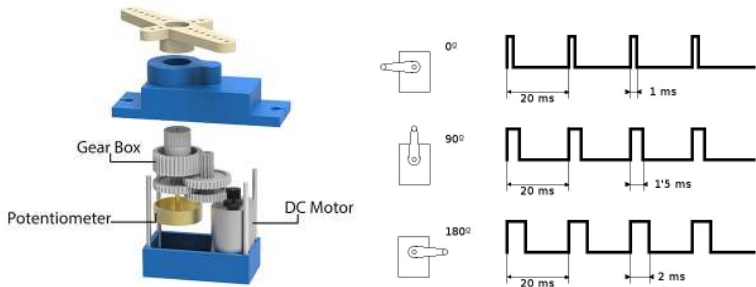
Many low cost servos used in robotics have a limited range of motion and can be moved to a specific position within that range.

The small and very inexpensive SG-90 servomotor (180° rotation, €3).



Actuators: Servomotor

The most common “hobby” servomotors use a potentiometer to “detect” their position and can be controlled with PWM. For example a 1.5ms pulse on many servos moves to the 90° position.



Images from <https://thestempedia.com/tutorials/what-is-a-servo-motor/amp/> and <https://www.makerelectronico.com/tutorial-servomotor-0-180-arduino/>

Actuators: Servomotor

The main specifications of interest are:

- Rotation range
- Speed - usually time to rotate 60° .
- Torque - kilogram/centimeters (2 kg/cm means it can support 2kg 1cm from the shaft, 1kg 2cm from the shaft, 4kg .5cm from the shaft etc.)
- Size

For example the analog SG-90 by TowerPro:

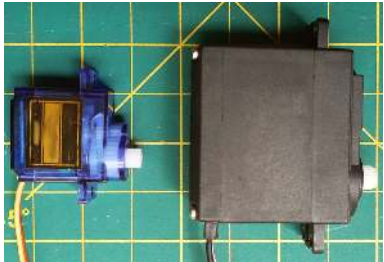
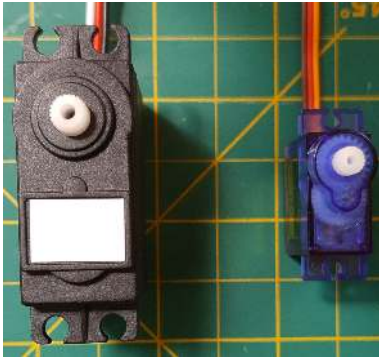
- Weight: 9g
- Dimension: 23x12.2x29mm
- Stall torque: 1.8kg/cm (4.8v)
- Rotation: about 180° .
- Operating speed: 0.12 sec/ 60° (4.8v)
- Operating voltage: 4.8v
- Temperature range: 0C-55C

Image and specs from <https://www.towerpro.com.tw/product/sg90-7/>



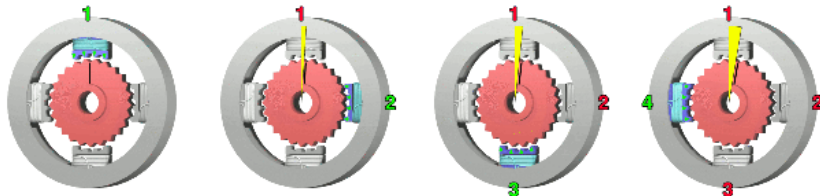
Actuators: Servomotor

Servos can be modified to allow continuous motion (by removing the potentiometer). This modification removes the possibility to move to a specific position but allows the control of direction and velocity. (You can also buy a continuous rotation servo.)



Actuators: Stepper Motor

A stepper motor (or step motor) is a brushless DC electric motor that divides a full rotation into a number of equal steps. The motor's position can then be commanded to move and hold at one of these steps without any feedback sensor (an open-loop controller), as long as the motor is carefully sized to the application. (Wikipedia)



Images from <http://en.wikipedia.org/wiki/File:StepperMotor.gif>

Actuators: Stepper Motor

Stepper motor control:

- The sequence of the coil pulses decides the direction in which the motor spins.
- The frequency of the coil pulses decides the speed of the motor's rotation.
- The number of coil pulses determines how much the motor will rotate.

While servomotors do not consume power when not moving, stepper motors do.

Actuators: Stepper Motor

Common uses of stepper motors:

- 3D printing, CNC cutting
- Position laser of CD or DVD
- Video camera zoom

Key specs:

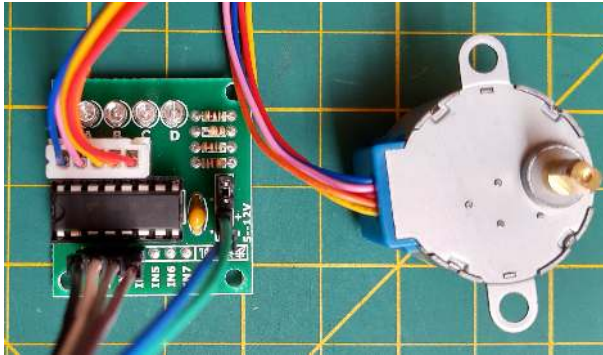
- Step angle or steps/revolution
- Holding torque (when powered)
- Shaft type
- Voltage and current rating
- Size



Image from https://commons.wikimedia.org/wiki/File:Nema_17_Stepper_Motor.jpg

Actuators: Stepper Motor

A 28BYJ-48 stepper motor with a ULN2003 driver. The motor has 32 steps per revolution in full step mode ($1 \text{ step} = 11.25^\circ$) but with its built in approx. $1/64$ gear reduction you get about 2048 steps per revolution (often closer to 2038). (About €4 for both)



Actuators: DC Motors

- Brushed motors

- economical
- can be noisy
- can cause electrical noise in circuitry

- Servomotors

- precise position and velocity control
- can actively control their position
- more expensive and complicated than steppers
- inexpensive models often have a limited range of motion

- Brushless motors

- require a controller
- compared with brushed motors they are: more expensive, more efficient, more durable and quieter

- Stepper motors

- precise control over angular movement and speed
- require a controller
- can hold their positions
- torque lowers as speed increases
- constant power consumption (independent of the load)

Actuators: DC Motors

A simple comparison:

	Characteristics		Control	
	Velocity	Torque	Positioning	Velocity
Brushed DC	High	Low	Bad	Bad
Geared down brushed DC	Med.	High	Bad	Bad
Brushless DC	Very High	Low	Bad	OK
Servo	Low	High	Good	Good
Servo cont. rotation	Low	High	Bad	OK
Stepper	Med.	Med.	Good	Good

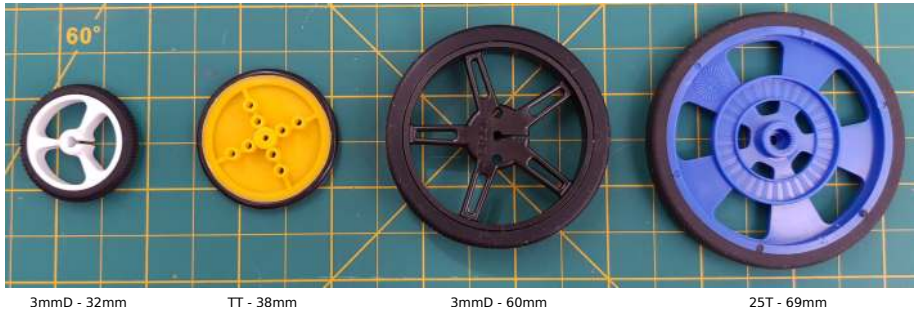
Note, adding an encoder to the motor resolves (in most cases) the problems related to control.

Table from <https://www.luisllamas.es/tipos-motores-rotativos-proyectos-arduino/>

Effectors: Wheels

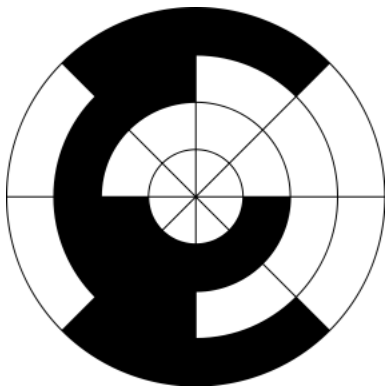
The wheels need to be compatible with the motor's shaft.
(Adapters can also be used.)

The diameter of the wheel and the RPM of the motor will determine the robot's speed.



Sensors - Absolute Encoder

A 3 bit optical encoder disc



Decimal	Binary	Gray Code
0	000	000
1	001	001
2	010	011
3	011	010
4	100	110
5	101	111
6	110	101
7	111	100

Image from [http://en.wikipedia.org/wiki/File:Encoder_Disc_\(3-Bit\).svg](http://en.wikipedia.org/wiki/File:Encoder_Disc_(3-Bit).svg)

Sensors - Absolute Encoder

A 13 bit optical encoder

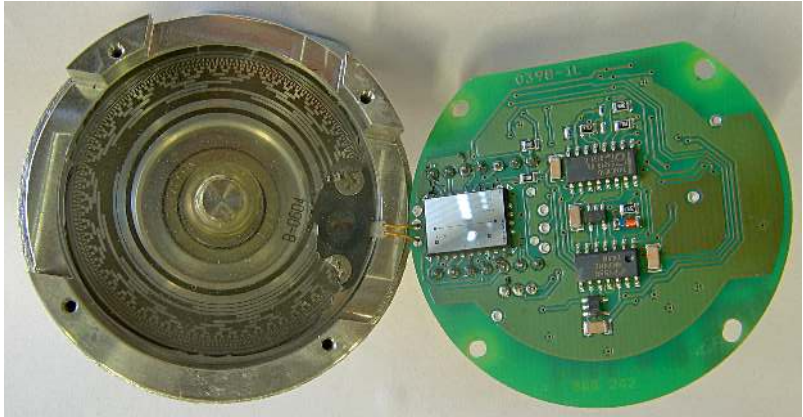


Image from http://en.wikipedia.org/wiki/File:Gray_code_rotary_encoder_13-track_opened.jpg

Types of Range Sensors

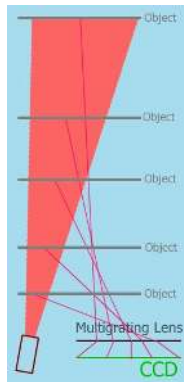
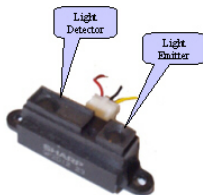
Given the importance of range sensors in many of the basic abilities of an intelligent robot, we will look at a number of different technologies used to construct them:

- Triangulation
- Structured light
- Time of flight

Types of Range Sensors - Triangulation (IR)

An example, Sharp IR Range Detectors.

They emit a pulse of IR light and a CCD array receives the signal, calculates the angle and uses that angle to estimate the distance.



Images from <http://www.acroname.com/robotics/info/articles/sharp/sharp.html>,
<http://www.howmuchsnow.com/snow/proto1.html>, and
<http://www.societyofrobots.com/robotforum/index.php?topic=7991.0>

Types of Range Sensors - Triangulation (IR)

Both 1-bit and analog versions with different ranges are available (again, example from Sharp):

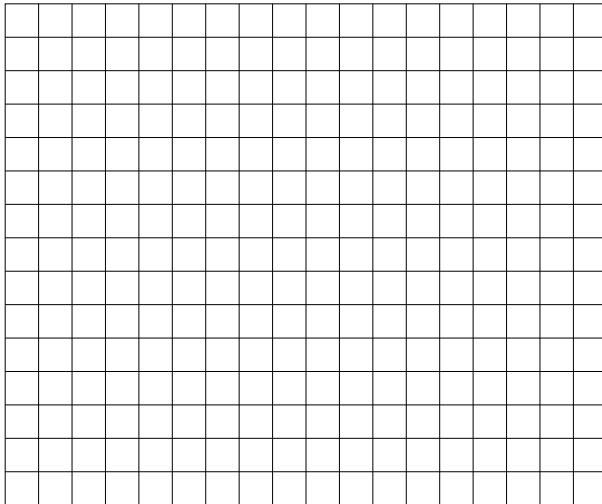
Distance Measuring Sensor Lineup

Sensor type	Output	Detected distance	Features	Model No.
PSD	1-bit digital output according to distance measuring	13 cm	1-bit digital output	GP2Y0D413K0F
		24 cm	1-bit digital output	GP2Y0D21YK0F
		80 cm	1-bit digital output	GP2Y0D02YK0F
	Analog voltage output according to distance measuring	1.5 to 15 cm	Analog output	GP2Y0AF15 series
		2 to 15 cm	Analog output	GP2Y0A51SK0F
		4 to 30 cm	Analog output	GP2Y0A41SK0F / GP2Y0AF30 series
		10 to 80 cm	Analog output	GP2Y0A21YK0F
		20 to 150 cm	Analog output	GP2Y0A02YK0F
		100 to 550 cm	Analog output	GP2Y0A710K0F

Image taken from <https://global.sharp/products/device/lineup/selection/opto/haca/diagram.html>

Types of Range Sensors - Structured Light

The basic idea: measure distances by analyzing the distortions in a known pattern.



Types of Range Sensors - Structured Light (Kinect)

The Kinect uses an infrared laser projector along with CMOS sensor (monochrome). It works in the range of about 0.8m - 4m.



From left to right: the projector, the RGB camera, the IR camera.

Image from <http://en.wikipedia.org/wiki/File:Xbox-360-Kinect-Standalone.png>

Types of Range Sensors - Structured Light (Kinect)

The Kinect uses a 'speckle' pattern of IR light. The system is based upon Light Coding Technology by PrimeSense.

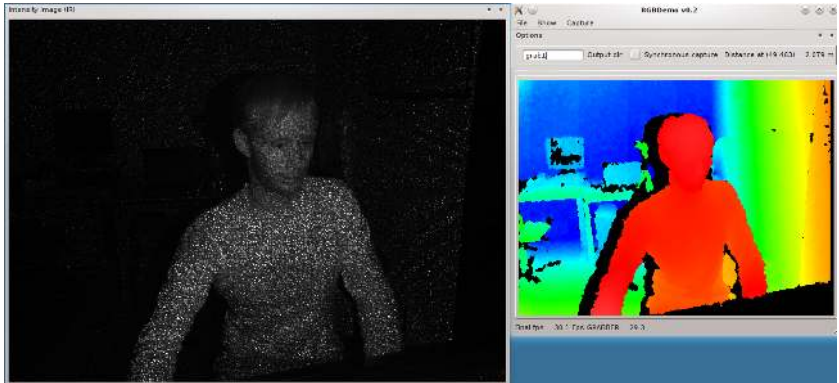


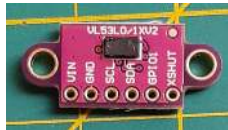
Image from

<http://nicolas.burrus.name/index.php/Research/KinectRgbDemoV4?from=Research.KinectRgbDemo>

Types of Range Sensors - Time of Flight

Time of Flight sensors measure the amount of time required for a wave, particle, ... to travel through a medium.

We have already talked about sonar sensors which are of this type (below left, HC-SR04, 2-400cm, 30° FOV). Another good example is the VL53L0X (below right, 2-125cm or 4-200cm (long range mode), 25° FOV).



Types of Range Sensors - Time of Flight (Modulation)

These sensors measure distance by looking at the phase shift of the wave. This shift is proportional to the time of flight (round trip), and from this we can calculate the distance.

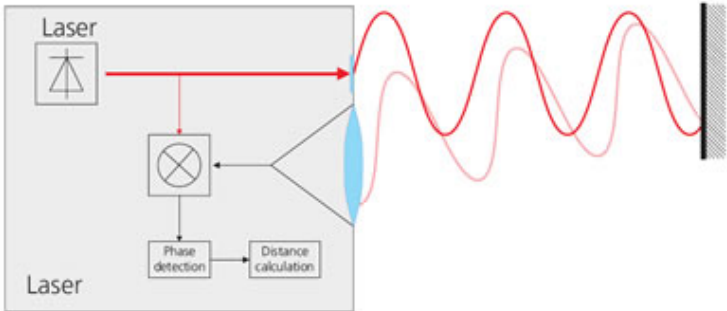


Image from <http://www.metrologyresource.com/laser-sensor-MRL3.php>

Types of Range Sensors - Time of Flight (Modulation)

Hokuyo URG-04LX Scanning Laser Rangefinder: ranges from 20mm to 4m (1mm resolution) in a 240° arc (0.36° angular resolution, 100 ms each scan)



Image and text from <http://www.robotshop.com/ca/en/hokuyo-urg-04lx-laser-rangefinder.html>

Types of Range Sensors - Time of Flight (Modulation)

Hokuyo URG-04LX Scanning Laser Rangefinder: ranges from 20mm to 4m (1mm resolution) in a 240° arc (0.36° angular resolution, 100 ms each scan)

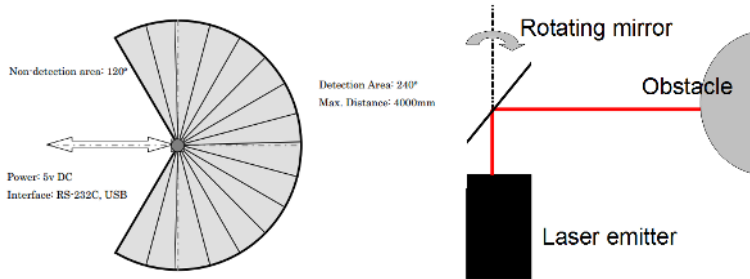


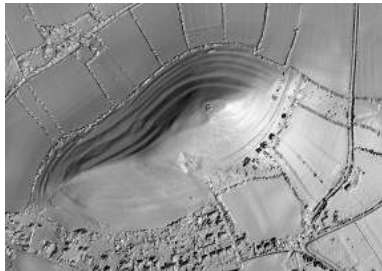
Image from <http://www.metrologyresource.com/laser-sensor-MRL3.php>

<http://www.generationrobots.com/en/content/52-localize-with-a-hokuyo-laser-range-finder>

What is Lidar?

A Lidar (LIght Detection And Ranging) Sensor is a laser based device which measures the round-trip time of light to calculate distances.

The use of the term is very broad and includes many applications: topography, measuring vegetation, atmospheric measurements, and speed guns. It also includes many different kinds of techniques (airborne, terrestrial,...) and technology (mechanical, solid state ...).



The Hokuyo URG-04LX is an example of one.

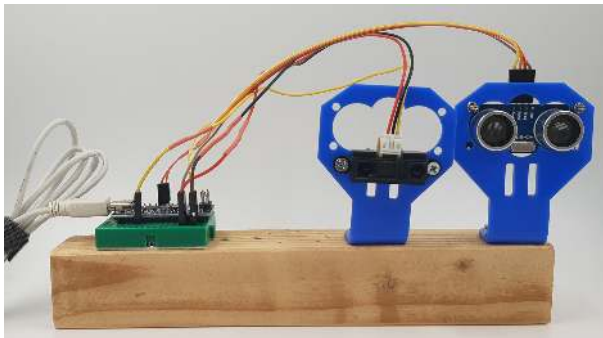
Image from <https://commons.wikimedia.org/wiki/File:Glastonbury-tor-ST5129638553.png>

A Quick Demo of Range Sensors (1)

Left: IR - Sharp GP 2Y0A21YK 0F (10-80cm)

Right: Sonar - HC-SR04 (2-400cm, 30° FOV)

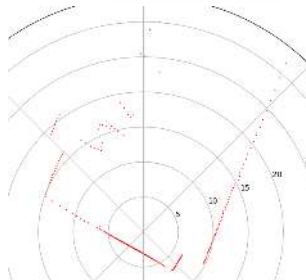
(Both connected to an Arduino Nano.)



A Quick Demo of Range Sensors (2)

The YDLIDAR X2 (laser (class 1) triangulation):

- Range: .12m-8m
- Scan Angle: 360°
- Scan Resolution: $.6^\circ - .96^\circ$
- Scan Frequency: 5-8Hz



Specifications and sensor image from <https://www.ydlidar.com/products/view/6.html>

Sensors: Video

The ESP32-CAM comes with an OV2640 camera, a microSD slot, WIFI (2.4 GHz), Bluetooth, a flash (LED), and pins for other uses.

You can run a little web server on the ESP32 and serve JPG on request from the camera over WIFI.

Unlike the Arduino, the ESP32-CAM does not include a USB to serial converter chip so you need a FTDI and jumper cables to program it.

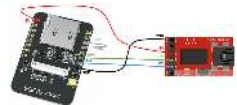


Image from <https://randomnerdtutorials.com/program-upload-code-esp32-cam/>

Communication: Bluetooth

It is easy to setup a serial connection over Bluetooth with an Arduino to send and receive information from the robot.

Some microcontrollers include Bluetooth modules, the most common Arduinos do not.

The HC-02 and the HC-06 (right). Both use classic Bluetooth, not BLE (Bluetooth Low Energy).

The HC-06 can only work in slave mode, the HC-05 can work in both master (can initiate connections) and slave mode.



Communication: WIFI

Again, some microcontrollers include WIFI modules, the most common Arduinos do not.

A WIFI shield is available for the Arduino. You can setup a little web server or send/receive UDP packets



Image from https://commons.wikimedia.org/wiki/File:WiFi_Shield_for_Arduino.jpg

Power: Batteries

You can power the Arduino via USB, its power jack (2.1mm center-positive plug), or the VIN pin. The jack accepts 7-12v DC and VIN accept 5-18v DC (however 7-12v DC is recommended), the Arduino has a regulator to produce the 5v it needs.

You can use the 5v pin of the Arduino to power other things (about 500mA max.). The L298N also has a regulator and a 5v output (again about 500mA max.)

Some battery options for a simple robot:

- 4 rechargeable AA (1.2v) (NiMH)
- 2 rechargeable 18650 (3.7v) (Li-Ion)
- 1 rechargeable 9v (NiMH or Li-Ion) - LOW mAh!



LiPo (and Li*) batteries can be dangerous, be careful!